



Fig. 3: Comparison of MPEL at 0.03-300MHz frequency range established as the obligatory standard for the general public of Russia versus similar values recommended by ICNIRP

the Institute of Electrical and Electronic Engineers (IEEE). These limit the specific absorption rate (SAR) to 1.6 watts per kilogram for part of the body (head exposure), 0.08 watt per kilogram for the whole body, and 4 watts per kilogram for exposure to hands, wrists, feet and ankles.

The FCC studies are based on the studies conducted on animals in 1970s and 1980s. Based on recommendations of the IEEE, the SAR level of 4 watts per kilogram is adopted for mobile phones. An analysis made by the Environmental Protection Agency (EPA) concluded that biological effects occur at SAR levels of 1 watt per kilogram—one-fourth the SAR level chosen by the FCC. Radio frequency has shown to induce heating of tissues leading to behavioural changes in mice, rats and monkeys, indicating potentially adverse effects in human beings too.

Russian standards compared to ICNIRP

Studies on EMF's health effects were initiated in Russia in 1948. In the following years, comprehensive hygienic, clinical, physiological and experimental studies were carried out. The contemporary Russian electromagnetic safety standards (sanitary and

obligatorily taken into account while assessing the risk of EMF's negative effects on human health to develop Russian standards of electromagnetic safety.

It was also demonstrated that health effects of long-term exposure to EMF over many years may accumulate to cause degeneration of the central nervous system, leukaemia, brain tumours, cardiovascular diseases and hormonal changes later. EMF can be especially dangerous to children, pregnant women, people with central

Power density exposure limits for GSM1800 adopted by various countries

- Australia - 2 W/m²
- Austria, Salzburg City - 0.001 W/m² (precautionary limit)
- Belgium (ex Wallonia) - 1.2 W/m²
- Canada - 3 W/m²
- CSSR, Belgium, Luxembourg - 0.24 W/m²
- India - 9.2 W/m² (as per ICNIRP recommendations)
- Italy, in areas with duration greater than four hours - 0.095 W/m²
- Italy, in sensitive areas - 0.025 W/m²
- New South Wales, Australia - 0.00001 W/m²
- New Zealand - 0.5 W/m²
- Poland, China, Italy and Paris - 0.1 W/m²
- Switzerland - 0.095 W/m²
- USA - 10 W/m² (as per FCC OET-65 Public Exposure Guidelines at 1800 MHz)

epidemiological norms and regulations, SanPIN) are developed based upon the analysis of these study results. These studies including chronic (long-term) EMF exposure have given the opportunity to identify the most sensitive systems of the human body—nervous, endocrine, immune, cardiovascular and reproductive systems. These critical system reactions were

nervous, hormonal and cardiovascular system diseases and people with allergies.

The basic requirement for permissible EMF levels established in Russia is the absence of even transient homeostasis disorder (including reproductive function) as well as the tension of protective and adaptive compensatory mechanisms—both at the nearest and late periods of time (which is different from EMF regulation system adopted in Western Europe and the US).

Thus maximum permissible levels of EMF established by Russian national standards at the 0Hz-300GHz frequency range are the strictest worldwide and provide the maximum protection to humans under EMF exposure.

How concerned is India

India does not follow many of the guidelines that are followed in the US. For example, in the US there should be no residential building within 50-metre radius of the cellphone tower. But in India, towers are created over lodges and multi-storied buildings. Cellphones might spread health hazards in epidemiological proportions. The types of diseases as per the research are due to the impact on cell growth in the human body and cell differentiation. Immune system, reproductive system and cardiovascular system are the physiological functions affected.

Since communication systems cannot be avoided, power levels should be reduced and guidelines strictly followed. Users should take precautions and manufacturers should be made to follow norms.

The Indian government has started a research programme for analysing the effects of mobile radiations on human health and environment after receiving a report from IIT professor Girishkumar. Its objectives are:

1. To study the standards followed by different agencies on mobile power limits
2. To compare the levels
3. To go through the historical